Springboard Virtual Internship 6.0

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College: Sona College of Technology

Branch: Computer Science and Engineering

Domain: Data Visualization

Project: Analysis of U.S. Natural Disaster Declarations

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1. Objective

The objective of this milestone is to analyze disaster incident types and understand their distribution across different regions. The analysis focuses on identifying frequently occurring disaster types and examining how they vary across states.

This milestone also aims to combine insights from previous analyses such as temporal trends and geographical patterns to provide a comprehensive understanding of disaster behavior.

The final goal is to extract meaningful insights regarding disaster frequency, regional variations, and overall patterns.

2. Problem Formulation

Analysis Type

This milestone includes:

- Exploratory Data Analysis (EDA)
- Comparative Analysis
- Categorical Data Analysis

Key Questions

The analysis is guided by the following questions:

- Which disaster types occur most frequently?
- Which states are more affected by specific disaster types?
- What patterns can be observed across different regions?
- What overall insights can be derived from the data?

3. Dataset Preparation

Input Dataset

The dataset used for this analysis is:
usnd_cleaned.csv
This dataset has been preprocessed in earlier milestones, including cleaning, feature extraction, and aggregation.

Columns Used

The following columns were used:

- incidentType – Type of disaster
- state – State where disaster occurred
- year – Year of occurrence
- month – Month of occurrence

Data Validation

Before analysis, the dataset was checked to ensure:

- No inconsistencies in incident types
- Proper distribution across states
- Availability of required columns

These steps ensured the dataset is reliable for analysis.

4. Data Aggregation

4.1 Incident Type Aggregation

The dataset was grouped based on incident type to calculate total occurrences.
This helps identify:

- Most frequent disasters

- Overall distribution of disaster types

4.2 State and Incident Type Aggregation

The dataset was grouped using:

- State
- Incident Type

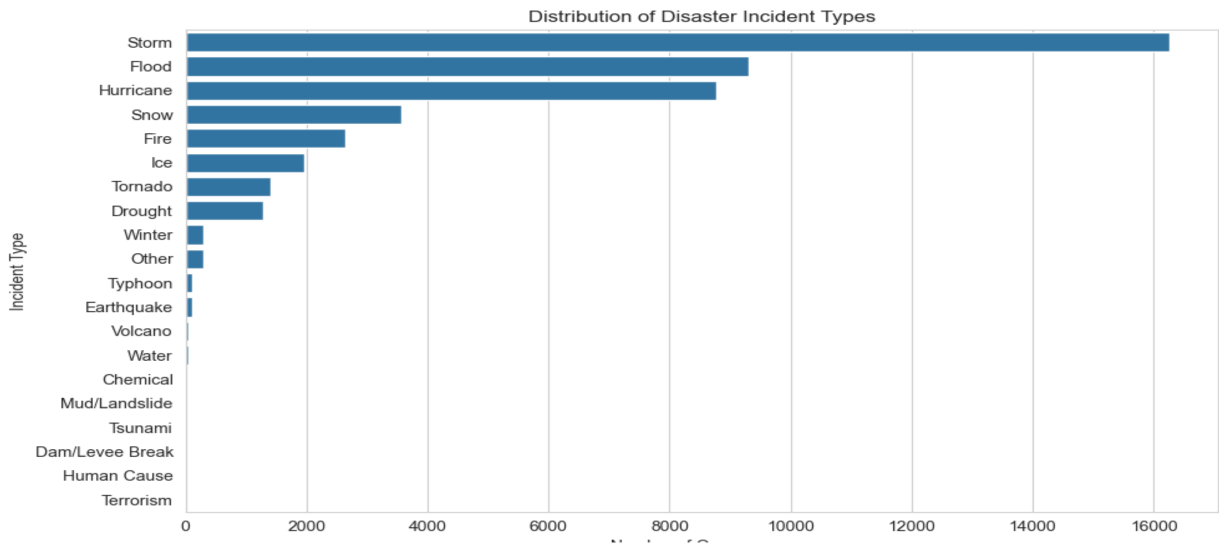
This helps in understanding:

- Regional disaster patterns
- Dominant disaster types in specific states

5. Visualizations

5.1 Distribution of Incident Types

A bar chart was created to visualize the frequency of each disaster type.
Interpretation:
The chart shows that certain disaster types occur more frequently than others. Weather-related disasters such as storms and floods dominate the dataset.

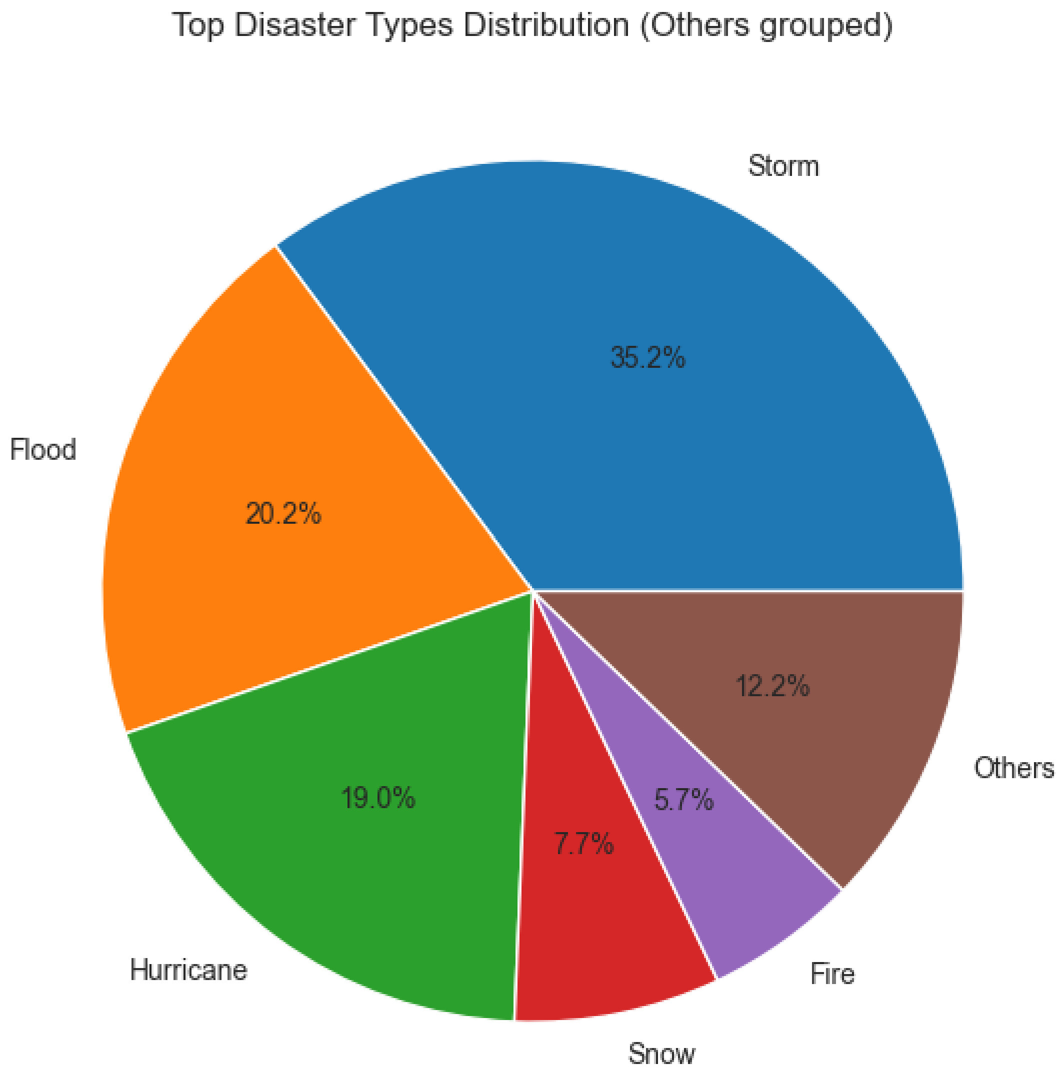


5.2 Pie Chart of Incident Types

A pie chart was used to represent the proportion of major disaster types.

Interpretation:

This visualization provides a clear view of how different disaster types contribute to the total occurrences. Major categories occupy a larger share compared to less frequent ones.

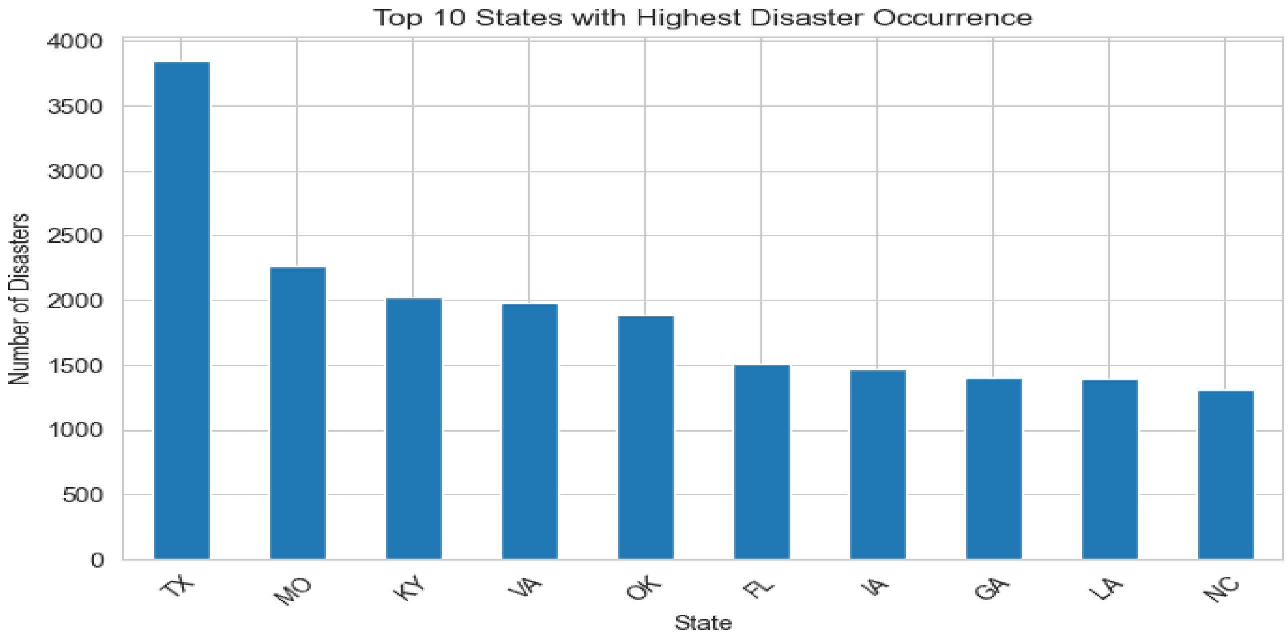


5.3 State vs Incident Type Analysis

A grouped analysis was performed to observe how disaster types vary across states.

Interpretation:

Different states show different disaster patterns. Some states experience higher frequency of certain disaster types, indicating regional vulnerability.

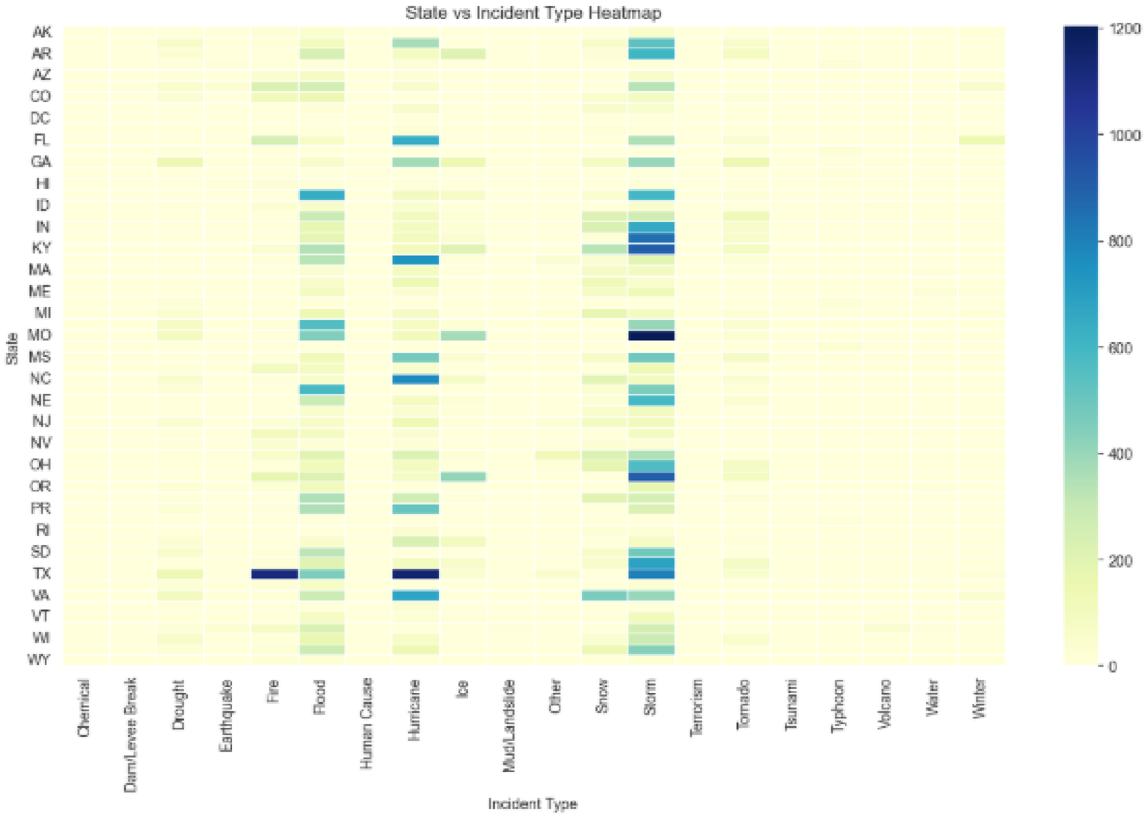


5.4 Heatmap of Disaster Distribution

A heatmap was created to visualize disaster frequency across states and incident types.

Interpretation:

The heatmap highlights regions with higher disaster occurrences. Darker shades represent higher frequency, helping identify disaster hotspots.



5.5 Assistance Program Analysis

The dataset used for this analysis did not include assistance-related columns such as ihProgramDeclared and paProgramDeclared.
Therefore, this part of analysis could not be performed.

6. Tools and Technologies Used

Data Processing

- Pandas – Data handling and aggregation

Visualization

- Matplotlib – Basic plotting
- Seaborn – Advanced visualization (heatmaps)

7. Key Insights

- Disaster events such as storms and floods are the most frequent.
- Disaster occurrence varies significantly across different states.
- Certain regions are more prone to specific disaster types.
- Clear patterns are observed when combining geographic and incident analysis.
- Visualization techniques help in understanding disaster trends effectively.

8. Conclusion

In this milestone, disaster incident types were analyzed to understand their distribution and regional patterns.
The analysis revealed that disaster occurrences are not uniform and vary based on location and type. Certain disaster categories dominate the dataset, while others occur less frequently.
By combining insights from previous milestones, a comprehensive understanding of disaster trends was achieved. This analysis can be useful for disaster preparedness and planning.

9. Dashboard Implementation Details

The Disaster Analytics Dashboard was developed using **Streamlit**, an open-source Python framework for building interactive web applications. The dashboard provides a user-friendly interface to explore disaster data and gain insights through visualizations.

Tools and Technologies Used:

- Python
- Streamlit
- Pandas (for data processing)
- Matplotlib / Seaborn / Plotly (for visualization)
- GitHub (for version control)
- Streamlit Community Cloud (for deployment)

Development Process:

- The dataset was first cleaned and preprocessed using Pandas.
- Missing values were handled, and relevant columns were selected.
- Multiple visualizations such as bar charts, pie charts, and trend graphs were created to analyze disaster frequency, types, and regional distribution.
- The dashboard was structured into sections like:
    - Data Overview
    - Missing Values Summary
    - Key Insights
    - Visual Analysis

Deployment:

The dashboard was deployed using Streamlit Cloud by connecting the GitHub repository containing the project files.

Dashboard URL:

https://us-natural-disaster-dashboard.streamlit.app